

# NIS2 ARTICLE 21

## COMPLIANCE SCAN

Prepared for

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Audit Date 2026-06-05

COMPLIANCE SCORE

**45**

/ 100

Critical findings. Immediate action required before  
audit.



**NOT AUDIT READY — Resolve findings before regulatory inspection**

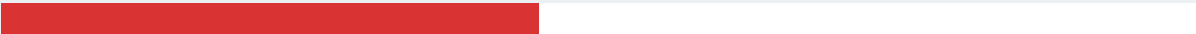
# EXECUTIVE SUMMARY

Compliance Score

45 / 100 — CRITICAL ISSUES

Critical findings. Immediate action required before

audit.



Clean Pass

1

Pass / Warnings

0

Failed

7

Inconclusive

0

Critical Findings

1

High Findings

4

Medium Findings

4

NOT AUDIT READY — Findings must be resolved before inspection

# DETAILED FINDINGS

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## Article 21(2)(a) — Network Security

**STATUS: FAIL**

Score: 55/100

*Port scan complete: 1 host(s) scanned*

LOW

Unencrypted HTTP (port 80) open with no HTTPS on

127.0.0.1

*Redirect HTTP traffic to HTTPS. Ensure TLS is enabled for all web services.*

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## Article 21(2)(e) — Patch Management

**STATUS: FAIL**

Score: 0/100

*Targets: 127.0.0.1*

*Automatic security upgrades enabled*

*OS: Ubuntu*

CRITICAL

No apt update history found

*Run 'apt update && apt upgrade' to install all available security updates.*

HIGH

10 pending security update(s)

*Apply immediately: 'apt-get upgrade'.*

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## Article 21(2)(i) — Access Control

STATUS: FAIL

Score: 55/100

Targets: 127.0.0.1

Privileged users: joshua

Root SSH login is disabled

Password policy configured

OS: Ubuntu

MEDIUM

SSH password authentication is enabled

Set 'PasswordAuthentication no' — use

key-based auth only.

MEDIUM

MaxAuthTries is 6 (recommended 4)

Set 'MaxAuthTries 4' in sshd\_config to limit

brute-force attempts.

MEDIUM

No SSH account lockout policy

(pam\_faillock/tally2) configured

Configure pam\_faillock in /etc/pam.d/common-auth

to lock after 5 failed attempts.

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## Article 21(2)(c) — Backup & Recovery

STATUS: FAIL

Score: 20/100

Targets: 127.0.0.1

Backup timers active: Sat 2026-06-06 00:00:00 EAT 5h

20min Fri 2026-06-05 09:18:30 EAT - dpkg-db-backup.timer

dpkg-db-backup.service

*Recent backup files (< 7 days):*

*/var/backups/dpkg.status.1.gz*

*/var/backups/apt.extended\_states.1.gz*

*/var/backups/alternatives.tar.0*

*/var/backups/apt.extended\_states.0*

*/var/backups/dpkg.status.0*

*Backup system detected and operational*

**HIGH**

No restore test evidence found in the last 90  
days

*Perform a restore test and document the result.*

*Establish quarterly restore testing.*

## Article 21(2)(f) — Log Management

**STATUS: FAIL**

Score: 55/100

Targets: 127.0.0.1

*rsyslog is running*

*systemd-journald is running*

*Log retention meets 90-day NIS2 requirement*

*auditd active — system call audit trail enabled*

**LOW**

No remote log forwarding (SIEM) configured

*Forward logs to a central SIEM/syslog server for*

*NIS2 compliance and incident response.*

## Article 21(2)(a) — Firewall & Network Defence

**STATUS: PASS**

Score: 100/100

Targets: 127.0.0.1

UFW firewall is active

iptables INPUT policy is DROP/REJECT (good)

## Article 21(2)(h) — Cryptography & TLS

**STATUS: FAIL**

Score: 55/100

Targets: 127.0.0.1

TLS 1.0/1.1 rejected — modern protocols only

SSH RSA host key is 3072 bits

SSH ECDSA host key is 256-bit

Ed25519 SSH host key present

**MEDIUM**

No disk encryption (LUKS) detected

Enable LUKS disk encryption for data at rest

protection. Especially important for /home and

/var/lib.

## Article 21(2)(j) — Multi-Factor Authentication

**STATUS: FAIL**

Score: 20/100

Targets: 127.0.0.1

fail2ban active — brute-force protection enabled

No NOPASSWD sudo entries — password required for

privilege escalation

**HIGH**

No MFA PAM module found (google-authenticator,

pam\_u2f, pam\_oath, Duo)

Install MFA: 'apt install

libpam-google-authenticator' and configure in

/etc/pam.d/sshd. Consider Duo Security or hardware

*tokens for privileged accounts.*

HIGH

SSH AuthenticationMethods not configured — MFA

not enforced for SSH

*Set 'AuthenticationMethods*

*publickey,keyboard-interactive' in sshd\_config to*

*require both key and MFA.*

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# REMEDIATION ROADMAP

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*Priority-ordered actions to achieve NIS2 Article*

*21 compliance:*

**CRITICAL**

[Article 21(2)(e)] No apt update history found

*Action: Run 'apt update && apt upgrade' to install  
all available security updates.*

**HIGH**

[Article 21(2)(e)] 10 pending security update(s)

*Action: Apply immediately: 'apt-get upgrade'.*

**HIGH**

[Article 21(2)(c)] No restore test evidence found  
in the last 90 days

*Action: Perform a restore test and document the  
result. Establish quarterly restore testing.*

**HIGH**

[Article 21(2)(j)] No MFA PAM module found  
(google-authenticator, pam\_u2f, pam\_oath, Duo)

*Action: Install MFA: 'apt install  
libpam-google-authenticator' and configure in  
/etc/pam.d/sshd. Consider Duo Security or hardware  
tokens for privileged accounts.*

**HIGH**

[Article 21(2)(j)] SSH AuthenticationMethods not

configured — MFA not enforced for SSH

*Action: Set 'AuthenticationMethods*

*publickey,keyboard-interactive' in sshd\_config to*

*require both key and MFA.*

**MEDIUM**

[Article 21(2)(i)] SSH password authentication is  
enabled

*Action: Set 'PasswordAuthentication no' — use*

*key-based auth only.*

**MEDIUM**

[Article 21(2)(i)] MaxAuthTries is 6 (recommended  
4)

*Action: Set 'MaxAuthTries 4' in sshd\_config to limit*

*brute-force attempts.*

**MEDIUM**

[Article 21(2)(i)] No SSH account lockout policy  
(pam\_faillock/tally2) configured

*Action: Configure pam\_faillock in*

*/etc/pam.d/common-auth to lock after 5 failed*

*attempts.*

**MEDIUM**

[Article 21(2)(h)] No disk encryption (LUKS)  
detected

*Action: Enable LUKS disk encryption for data at rest*

*protection. Especially important for /home and*

*/var/lib.*

**LOW**

[Article 21(2)(a)] Unencrypted HTTP (port 80)

open with no HTTPS on 127.0.0.1

*Action: Redirect HTTP traffic to HTTPS. Ensure TLS  
is enabled for all web services.*

LOW

[Article 21(2)(f)] No remote log forwarding

(SIEM) configured

*Action: Forward logs to a central SIEM/syslog server  
for NIS2 compliance and incident response.*